

Assignment 4: Vital Sign Report- Ansarah Mohammed

Vital Sign Chosen: Mean Arterial Pressure (MAP)

Mean Arterial Pressure, also known as MAP, is the average pressure in a person's arteries during one complete heartbeat cycle. It gives an overall idea of how well blood is being pushed through the body and whether enough pressure is available to supply organs and tissues with blood and oxygen. MAP is useful because it combines both systolic and diastolic blood pressure instead of looking at only one blood pressure value on its own (DeMers & Wachs, 2023). It can be calculated using the formula:

$$MAP = \frac{(SBP + 2 * DBP)}{3}$$

In this formula, SBP means systolic blood pressure, which is the pressure when the heart contracts, while DBP means diastolic blood pressure, which is the pressure when the heart relaxes. Diastolic blood pressure is multiplied by 2 because the heart spends more time in the relaxation phase than in the contraction phase.

The clinical range for MAP is usually around 70–110 mmHg, values below 60 mmHg are considered dangerous, as it may mean that the body's vital organs are not receiving enough blood flow. Organs such as the brain, heart, and kidneys need a steady supply of blood and oxygen to function properly. If it remains too low, the patient may be at risk of organ damage, shock, or other serious complications (DeMers & Wachs, 2023; EBSCO, n.d.). NCBI also states that maintaining a minimum of about 60 mmHg is required for vital organ perfusion.

In triaging, this is important because it helps healthcare workers identify patients who may be unstable or in need of urgent medical attention. A low MAP may suggest poor circulation, dehydration, blood loss, trauma, sepsis, or shock. This means the patient may need to be assessed and treated quickly. A very high MAP can also be concerning because it may place extra strain on the heart and blood vessels, especially if the patient has symptoms such as chest pain, confusion, severe headache, or difficulty breathing.

Overall, MAP is an important vital sign because it helps show whether the body is receiving enough blood flow to support vital organs. In a triage setting, MAP can help determine whether a patient is stable or at risk of becoming critically ill. Abnormal MAP values should not be treated as simple outliers because they may represent real clinical issues that require attention.

References

- DeMers, D., & Wachs, D. (2023). *Physiology, mean arterial pressure*. In *StatPearls*. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK538226/>
- EBSCO. (2024). *Mean arterial pressure (MAP)*. EBSCO Research Starters. <https://www.ebsco.com/research-starters/health-and-medicine/mean-arterial-pressure-map>